# Group 5: Vision based eye snake robot localization

Background Reading Presentation

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# Project Overview

## Problem

Continuum surgical robots suffer from hysteresis → kinematics alone can't localize well

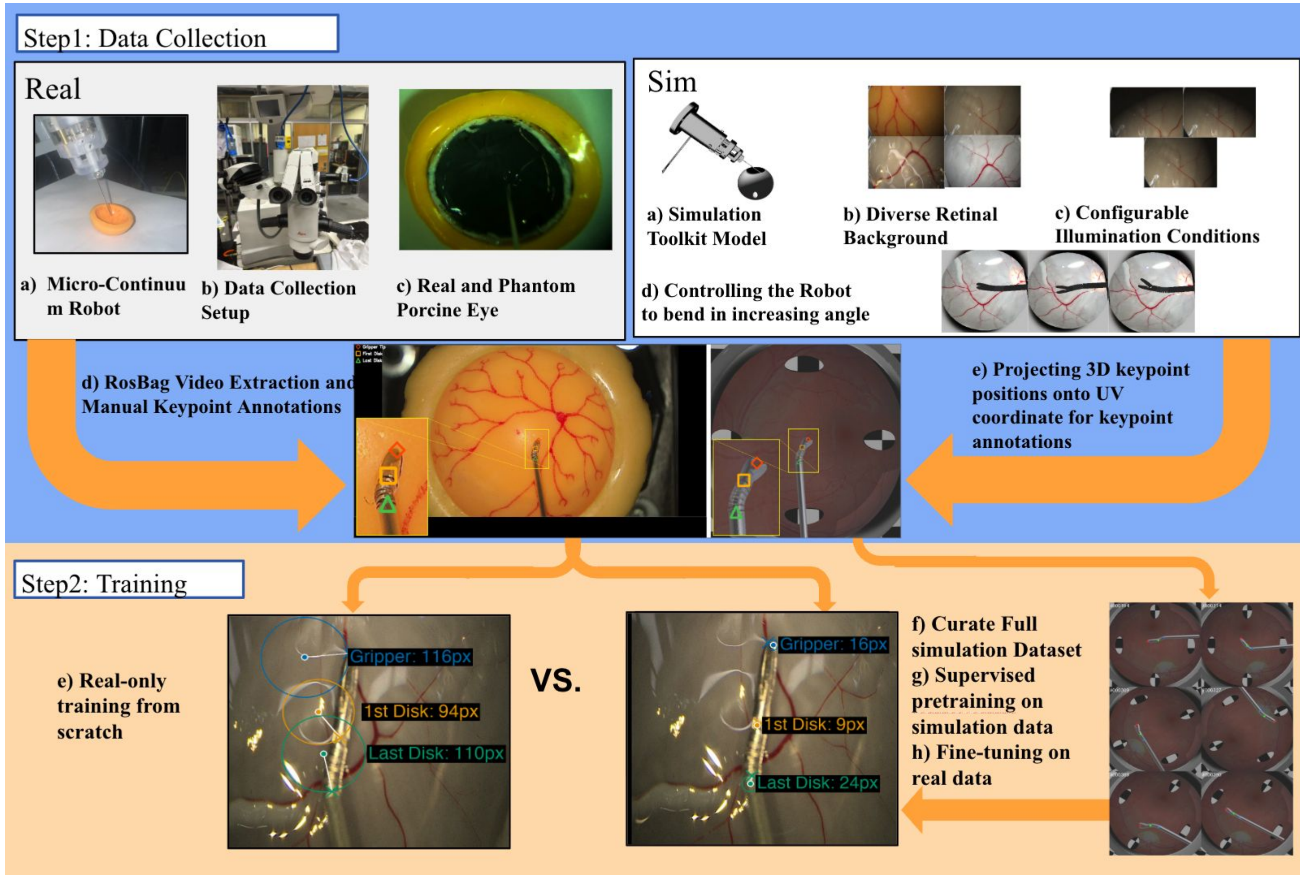
## Solution

Vision-based localization with sim-to-real transfer

We start with benchmarking all popular localization approach for exploration

### 20 Papers in 3 Categories:

1. Surgical Robots & Control
2. Simulation Frameworks
3. Current Localization methods
   - classical
   - heatmap
   - direct regression
   - integral regression
   - detection



# ➊ Surgical Robots & Control (8 papers)

## Why continuum robots? Why eye snake?

R. J. I. Webster and B. A. Jones, “Design and kinematic modeling of constant curvature continuum robots: A review,” The International Journal of Robotics Research, vol. 29, no. 13, pp. 1661–1683, 2010.

- continuum robots offer the potential to extend the benefits of robotics to applications that have not previously been approachable with robots. These include **working in unstructured or constrained environments and manipulating delicate objects** or those with precisely known shapes or locations.

## Why vision-based localization is needed

B. Y. Cho, D. S. Esser, J. Thompson, B. Thach, R. J. Webster, and A. Kuntz, “Accounting for hysteresis in the forward kinematics of nonlinearly-routed tendon-driven continuum robots via a learned deep decoder network,” IEEE Robotics and Automation Letters, vol. 9, no. 11, pp. 9263–9270, 2024.

- “Recent proposed learning-based methods have been shown to overcome some of these limitations **but do not account for hysteresis**, a significant source of error for these robots…”

## Other existing efforts on continuum robots

R. J. I. Webster and B. A. Jones, “Design and kinematic modeling of constant curvature continuum robots: A review,” The International Journal of Robotics Research, vol. 29, no. 13, pp. 1661–1683, 2010.

A. Uneri, M. A. Balicki, J. Handa, P. Gehlbach, R. H. Taylor, and I. Iordachita, “New steady-hand eye robot with micro-force sensing for vitreoretinal surgery,” in 2010 3rd IEEE RAS & EMBS International Conference on Biomedical Robotics and Biomechatronics (BioRob). Tokyo, Japan: IEEE, 2010, pp. 814–819

P. E. Dupont, J. Lock, B. Itkowitz, and E. Butler, “Design and control of concentric-tube robots,” IEEE Transactions on Robotics, vol. 26, no. 2, pp. 209–225, 2010.

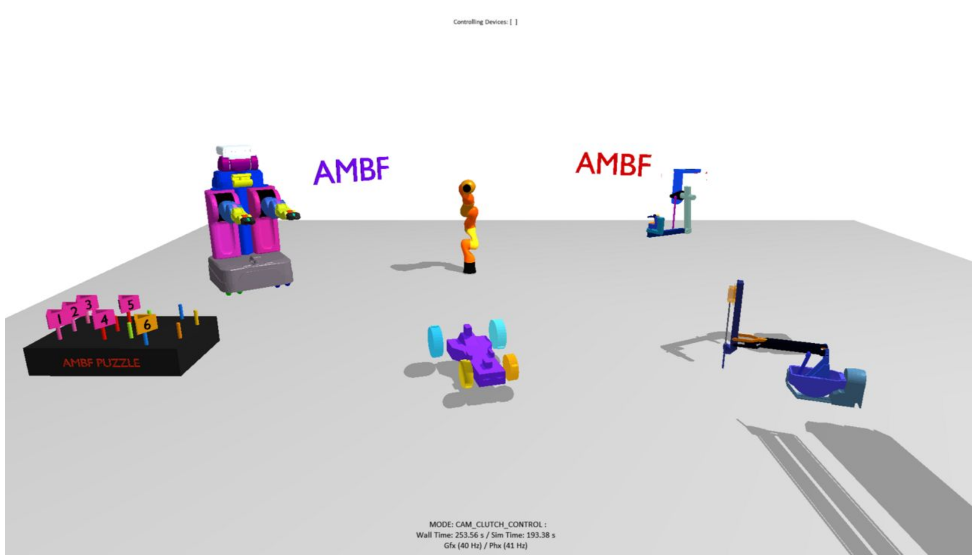
B. P. M. Yeung and T. Gourlay, “A technical review of flexible endoscopic multitasking platforms,” International Journal of Surgery, vol. 10, no. 7, pp. 345–354, 2012. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1743919112001069

J. Burgner-Kahrs, D. C. Rucker, and H. Choset, “Continuum robots for medical applications: A survey,” IEEE Transactions on Robotics, vol. 31, no. 6, pp. 1261–1280, 2015.

# ➋ Simulation Frameworks  (5 papers)

## We need simulation data because real data is scarce and hard to get

- **Munawar & Fischer, IROS 2019**  —  AMBF — our simulation platform (real-time physics + RGB-D)
  - AMBF can model any multi-body system with real-time physics, synchronized RGB-D capture, and ROS control




# ❷ Simulation Frameworks (5 papers)

## Other existing efforts on simulation framework

Z. Chen, A. Deguet, S. Vozar, A. Munawar, G. Fischer, and P. Kazanzides, “Interfacing the da vinci research kit (dvrk) with the robot operating system (ros),” in Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) Workshops, Seattle, WA, USA, 2015.

J. Xu, B. Li, B. Lu, Y.-H. Liu, Q. Dou, and P.-A. Heng, “Surrol: An open-source reinforcement learning centered and dvrk compatible platform for surgical robot learning,” in 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2021, pp. 1821– 1828.

# Classical Baselines

| Method | Ref | Original Paradigm | Full Citation |
|---|---|---|---|
| **Template Matching** | [16] | Classical (non-DL) | R. Brunelli, Template Matching Techniques in Computer Vision: Theory and Practice. John Wiley & Sons, 2009. |
| **Color Segmentation** | [17] | Classical (non-DL) | H.-D. Cheng, X. H. Jiang, Y. Sun, and J. Wang, "Color image segmentation: advances and prospects," Pattern Recognition, vol. 34, no. 12, pp. 2259–2281, 2001. |

*Template matching works by sliding a reference patch across the image and finding the location of maximum correlation.*
*Color segmentation isolates the robot tip by thresholding pixel values in a chosen color space.*

# Heatmap-Based Methods

| Method | Ref | Original Paradigm | Full Citation |
| --- | --- | --- | --- |
| **SimpleBaseline** | [18] | Heatmap + argmax | B. Xiao, H. Wu, and Y. Wei, "Simple baselines for human pose estimation and tracking," in Proc. ECCV, 2018. |
| **HRNet** | [19] | Heatmap + argmax | K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in Proc. CVPR, 2019, pp. 5686–5696. |
| **Stacked Hourglass** | [20] | Heatmap + argmax | A. Newell, K. Yang, and J. Deng, "Stacked hourglass networks for human pose estimation," in Proc. ECCV, Springer, 2016, pp. 483–499. |
| **CenterNet** | [21] | Detection + heatmap | X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," in Proc. ICCV, 2019. |
| **U-Net** | [22] | Heatmap + argmax | O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in Proc. MICCAI, 2015. |

*Heatmap-based methods are the dominant paradigm in human pose estimation and form the largest category in our benchmark. Rather than predicting coordinates directly, these methods output a spatial probability map for each keypoint, where the peak of the map indicates the predicted location.*

# Direct Regression Methods

| Method | Ref | Original Paradigm | Full Citation |
|---|---|---|---|
| **ResNet Direct** | [23] | Direct regression | K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. CVPR, 2016, pp. 770–778. |
| **ViT Direct** | [24] | Direct regression | A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in Proc. ICLR, 2021. |

*Direct regression methods take the most straightforward approach to keypoint localization: a backbone network encodes the full image into a global feature vector, and a regression head maps that vector directly to output coordinates.*

# Integral Regression Methods

*Method · Citation reference · Original paradigm · Full citation*

| Method | Ref | Original Paradigm | Full Citation |
|---|---|---|---|
| **Integral Regression** | [25] | Soft-argmax / integral | X. Sun, B. Xiao, F. Wei, S. Liang, and Y. Wei, "Integral human pose regression," in Proc. ECCV, 2018, pp. 529–545. |
| **ViTPose** | [26] | Heatmap + argmax | Y. Xu, J. Zhang, Q. Zhang, and D. Tao, "ViTPose: Simple vision transformer baselines for human pose estimation," Advances in NeurIPS, vol. 35, pp. 38571–38584, 2022. |
| **DINOv2 Regression** | [27] | Backbone only (no decoder) | M. Oquab et al., "DINOv2: Learning robust visual features without supervision," arXiv:2304.07193, 2023. |

*Integral regression methods were introduced to bridge the gap between heatmap-based and direct regression approaches. Like heatmap methods, they produce a spatial probability map for each keypoint, preserving the rich spatial structure of the image.*

Category 5 of 5

# Detection-Based Localization

| Method | Ref | Original Paradigm | Full Citation |
|---|---|---|---|
| **DETR** | [28] | Transformer detection | N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in Proc. ECCV, 2020. |

*Detection-based localization treats keypoint prediction as an object detection problem rather than a pose estimation problem.*